

Title

Effects of Electrical Noise Stimulation on Postural Control During Dynamic Balance Challenges in Parkinson Disease

Abstract

Gait instability in Parkinson disease reflects impaired sensorimotor integration and dysfunctional balance mechanisms. Electrical stimulation delivering sub-sensory-threshold noise may enhance sensory system responsiveness. We evaluated whether such stimulation improves postural stability during visually-challenging walking in 14 individuals with Parkinson disease. Participants completed walking trials within a virtual environment delivering dynamic visual perturbations, performed with and without individualized electrical noise stimulation. We measured center-of-mass sway, foot placement precision, and ankle stabilizing responses bilaterally. Peak center-of-mass lateral excursion was significantly increased during noise stimulation compared with the no-stimulation condition (mean difference 3.2 cm; $p=0.02$), indicating reduced postural stability. Paradoxically, the stimulation amplified rather than dampened the destabilizing effect of visual perturbations. Foot placement control metrics and ankle torque responses showed no significant between-condition differences. Exploratory analysis suggested the stimulation-induced increase in sensory noise overwhelmed compensatory balance mechanisms in this population. These findings indicate electrical noise stimulation may be counterproductive for gait stability in Parkinson disease.